INCH-POUND
MIL-DTL-24231/20E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/20D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTOR, RECEPTACLE, MULTI-PIN, TYPE V, 65-CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 3.

2. Material for receptacle body shall be manufactured from nickel-copper alloy in accordance with Class A, Form 1, hot-finished condition of QQ-N-281 with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy in accordance with Class A, Form 1, cold-drawn condition of QQ-N-281 with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset.

3. All dimensions are in inches.

4. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½degree.

5. These connectors are for maintenance purposes only.

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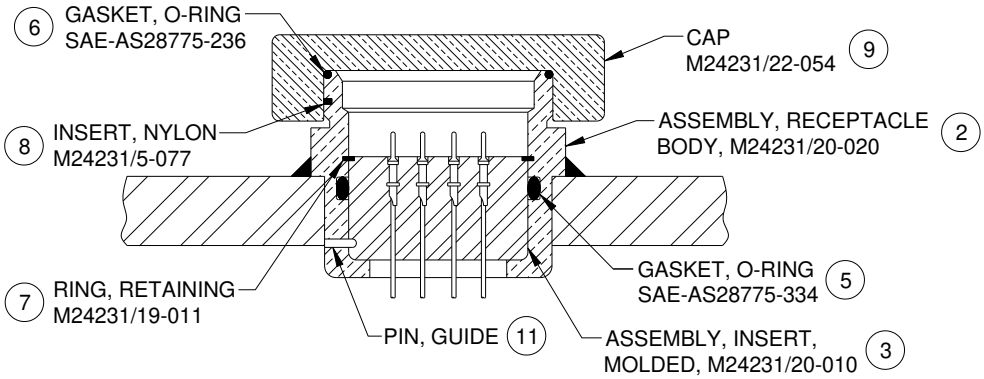


FIGURE 1. Connector, receptacle multi-pin – M24231/20-001 sym no. 1138.1. (1)

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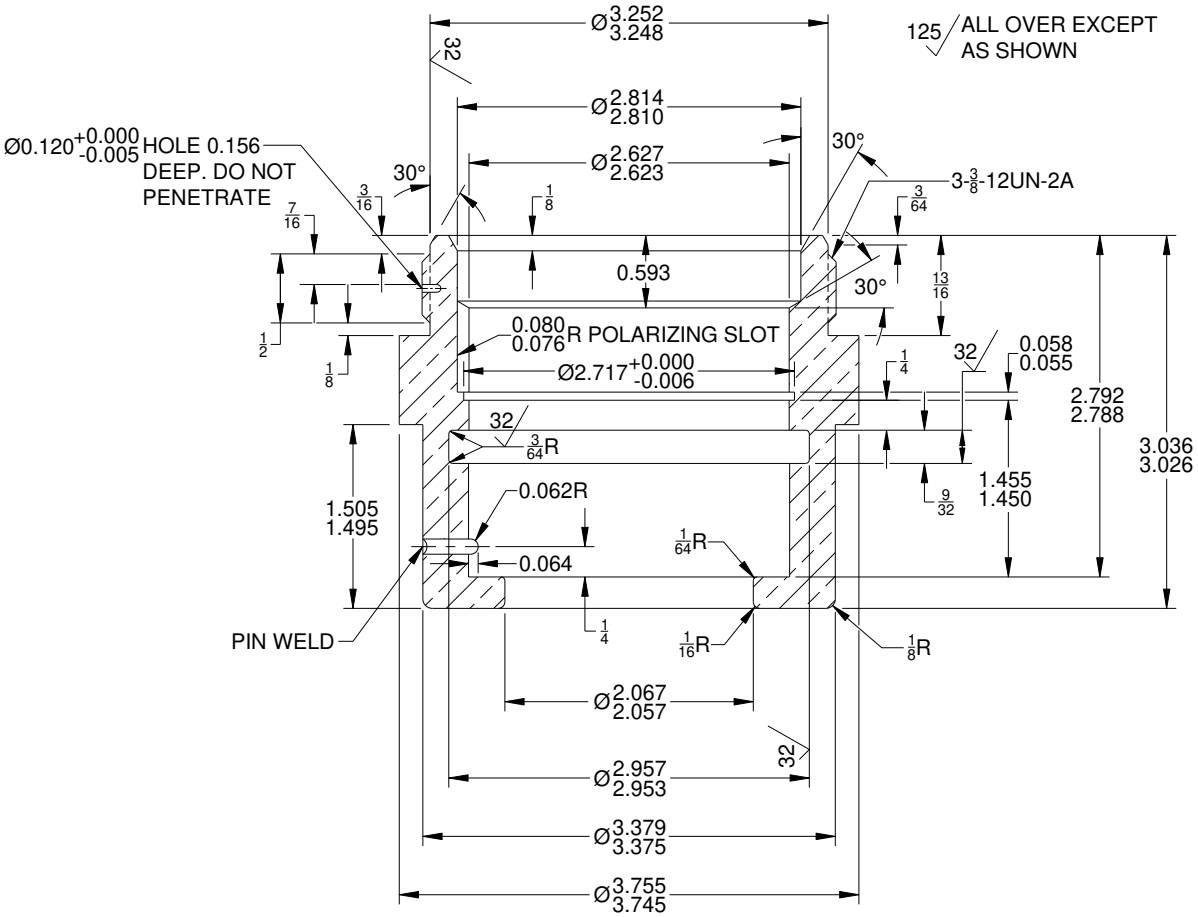


FIGURE 2. Assembly, receptacle body – M24231/20-020. (2)

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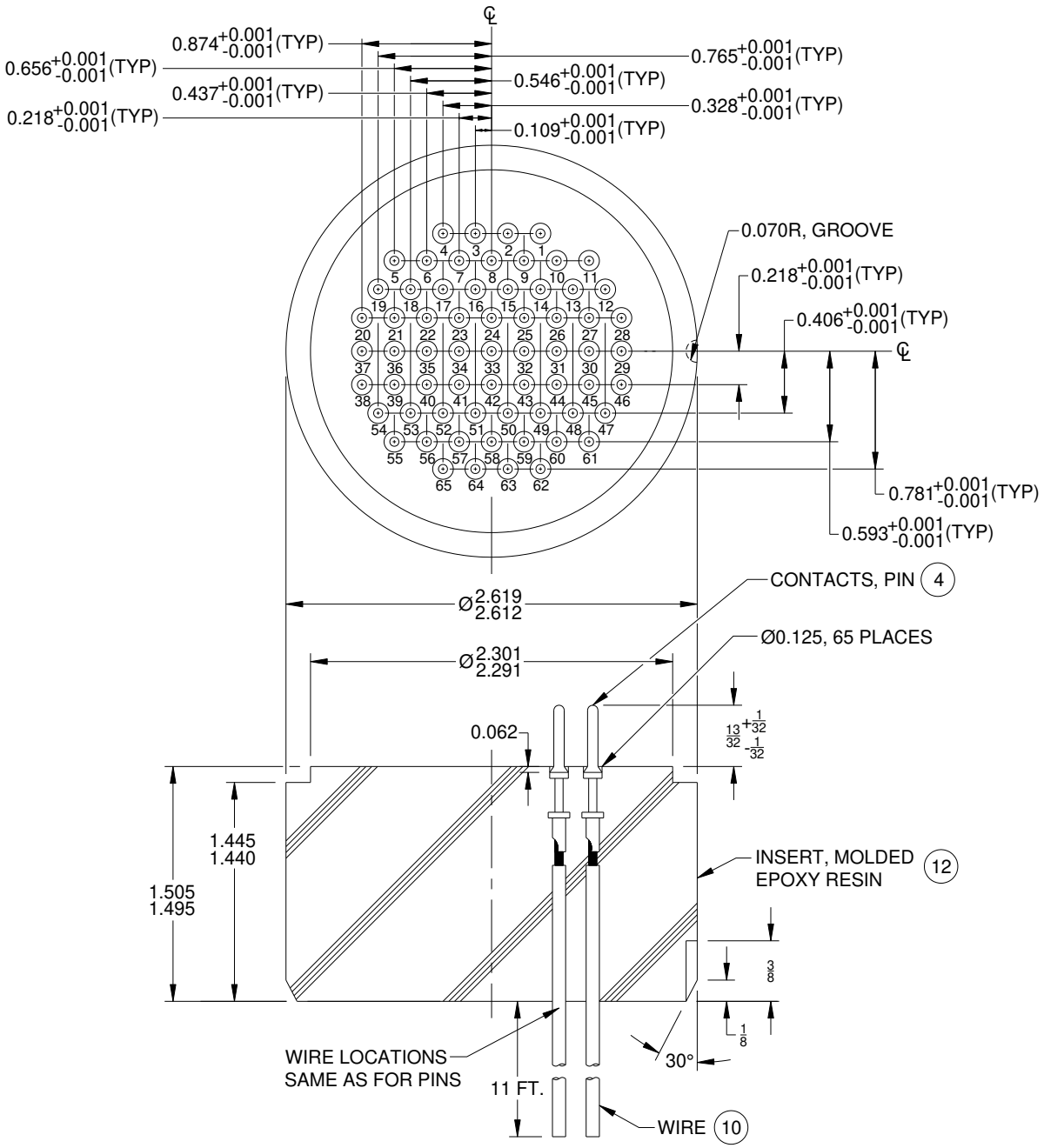


FIGURE 3. Assembly, molded insert – M24231/20-010. (3)

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/20- 3/	Material/specification	Remarks
		001		
1	Receptacle, 65-conductor symbol no. 1138.1	1	------	See figure 1.
2	Body, receptacle	1	Ni-Cu, Class A, QQ-N-281	See figure 2.
3	Insert assembly, molded	1	See items 04, 10, and 12.	See figure 3.
4	Pin, contact	65	MIL-DTL-24231	See figure 3.
5	Gasket, O-ring	1	SAE-AS28775-334 4/	Lubricate with silicone compound, SAE-AS8660; see figure 1.
6	Gasket, O-ring	1	SAE-AS28775-236 4/	Same as item 5.
7	Ring, retaining	1	M24231/19-011	See figure 1.
8	Insert	1	M24231/5-077	See figure 1.
9	Cap, pressure-proof	1	M24231/22-054	See figure 1.
10	Wire	As required	MIL-DTL-16878/36 BJE0	See figure 3.
11	Pin, guide, 0.125 diameter, length as required	1	Ni-Cu, Class A, QQ-N-281	See figure 1.
12	Epoxy resin, molded	As required	Cast epoxy resin compound 5/	See figure 3.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

5/ Arco Set 200, M.M. Reif and Co., Ivyland, PA, or PR610-TCF (Chockfast Orange), ITW Philadelphia Resins Corp., Montgomeryville, PA, or equal.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-069)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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